

Subjective Judging and the Host Effect at Japan's National Sports Festival: A 47-Prefecture Panel Analysis with a 2024-2025 Cross-Sectional Interaction Test

Running title: Subjective judging and the Kokutai host effect

Authors

Mizuki Shirai, MHS¹

¹ Specified Nonprofit Corporation Rehabilitation Collaboration, Suita, Osaka, Japan

Corresponding author: Mizuki Shirai, MHS, Specified Nonprofit Corporation Rehabilitation Collaboration, Suita, Osaka, Japan. Email: rehabilitation.collaboration@gmail.com. ORCID: 0009-0005-3615-0670.

Abstract

Background: Host-country advantage in international sport is well established, but Balmer, Nevill & Williams (2003) documented that the advantage is concentrated in *subjectively judged* events (boxing, gymnastics) rather than in objectively measured events (track, weightlifting). Japan's National Sports Festival (Kokutai) is an unusually clean natural experiment for this hypothesis — 47 prefectures rotating as hosts across 80 editions since 1946, with essentially identical eligibility rules — yet the leading Japanese-language predecessor (Funahashi et al. 2016) documented a robust host bonus without disaggregating by sport type. We test whether the Kokutai host bonus is larger in subjectively judged sports than in objectively measured ones.

Methods: We assembled a 47-prefecture panel of Kokutai overall rankings spanning 2012-2022 (n = 423 prefecture-years, main analysis) and reproduced Funahashi et al. (2016)'s 2003-2011 specification (n = 423) for direct comparability. As primary specifications we fitted (i) a pooled ordered logit on 1-8 finishing ranks censored at "outside top 8" with log-population and log-GDP as covariates (no prefecture or year fixed effects), (ii) a binary logit on top-1 (championship) status with the same covariate set, and (iii) a Csurilla-style staged analysis (host only → +population → +GDP → +prefecture FE → +year FE) — under which the two primary specifications above are numerically identical to the Csurilla M3 (+ log GDP) row, an equivalence we make explicit below. Because sport-by-prefecture-by-year score data are only publicly available for the two most recent editions (2024 Saga and 2025 Shiga), we tested the objective/subjective interaction in a stacked cross-section of 6,991 prefecture-sport-year cells (47 prefectures × 40-41 sports × 2 years × 2 trophies, prefecture-clustered SE). Sports were classified as objective (n = 17), subjective (n = 11), or semi-subjective/team (n = 13) following Balmer, Nevill & Williams (2003).

Results: In the 2012-2022 panel, host prefectures took the championship in 6 of 9 host-years (66.7%) and finished in the top 8 in 9 of 9 (100%) — a complete separation on top-8 status that was handled descriptively. The pooled ordered logit (with log-population and log-GDP controls) yielded a host coefficient of -13.73 (SE = 2.03, prefecture-clustered; $p < 0.001$; negative meaning better rank), and the pooled binary logit for top-1 (same controls) gave +9.09 (SE = 2.13, prefecture-clustered; $p < 0.001$). Our replication of Funahashi et al. (2016) for 2003-2011 recovered a host coefficient of +1,575.45 in total score (SE = 57.58; $p < 0.001$; $R^2 = 0.94$), within 5.9% of Funahashi’s reported +1,674.65 despite our specification omitting Funahashi’s socioeconomic controls (ESRI 2008 System of National Accounts (2008SNA) data are not available before 2011). Csurilla-style staged confounder controls did *not* attenuate the host coefficient; on the contrary, adding population and GDP strengthened it — the inverse of the ~45% attenuation Csurilla & Fertő (2023) report for the Olympics, and consistent with a host effect that is not confounded by prefecture size. In the 2024-2025 cross-section ($n = 6,991$), the host main effect was +16.60 points per sport (SE = 1.25; $p < 0.001$) and the **host $\times$ subjective interaction was +16.68 points (SE = 7.72; cluster-robust $p = 0.031$)** — the marginal host bonus in subjectively judged sports was roughly *double* the bonus in objectively measured sports. Because the cross-section contains only two treated prefectures (Saga 2024, Shiga 2025) out of 47, we complement the cluster-robust p-value with a wild-cluster bootstrap (Rademacher weights, restricted null, $B = 999$), which yielded $p = 0.155$; the interaction does not attain conventional significance under this few-treated-clusters correction and should be read as a first empirical anchor rather than as a decisive rejection of the null. Descriptively, the raw host-versus-nonhost gap was +37.9 points in subjective sports, +26.2 in semi-subjective/team, and +17.6 in objective — monotonically ordered as predicted by Balmer2003.

Conclusions: The Kokutai host bonus is not uniform across sport types. The subjective-versus-objective decomposition, absent from prior Japanese work, recovers a Balmer-2003-consistent pattern: the host advantage is concentrated in — and roughly doubled by — sports whose outcomes depend on judge scoring rather than a stopwatch or a tape measure. Because we can only estimate the interaction from two cross-sectional years (2 treated prefectures out of 47), the point estimate should be viewed as a first empirical anchor rather than a definitive magnitude, and the wild-cluster bootstrap that accounts for the few-treated-clusters structure does not reach conventional significance ($p = 0.155$). Nevertheless, the descriptive host-boost gap (+37.9 subjective vs. +17.6 objective, monotonically ordered as Balmer2003 predicts) and the log-outcome robustness check (interaction coefficient +0.293, same direction as the linear specification) both support the Balmer2003 direction, and the finding closes a specific gap left by Funahashi et al. (2016) and provides Japanese quantitative evidence complementing the qualitative critiques of Suetsugu (2024, 2025).

Keywords: Home advantage, host effect, subjective judging, National Sports Festival, Kokutai, panel data, Japan, ordered logit, Balmer 2003

Introduction

Does hosting a sporting event genuinely improve the host's performance, and if so, through what mechanism? At the international scale, the answer is unambiguously yes: host countries reliably win more medals than their non-host baseline predicts, an effect documented at every summer Olympics from 1896 forward.^{1,2} The mechanism, however, is contested. Six candidate pathways are typically invoked — host-country entry quotas, training-venue familiarity, home-crowd support, budget uplift, jet-lag/climate advantage, and, most contentiously, **subjective-judging bias in favor of the host** — but no single mechanism has been uniquely identified in any single dataset.

Balmer, Nevill & Williams (2003) proposed a diagnostic decomposition.³ If home advantage were driven by facility familiarity, budget, or crowd support, it should appear roughly uniformly across all Olympic sports. If, on the other hand, referee or judge bias were a load-bearing mechanism, home advantage should be concentrated in sports where the outcome depends on human scoring rather than an objective clock or tape measure. Comparing five event families across a century of summer Olympic data, Balmer2003 found that home advantage is indeed disproportionately concentrated in subjectively judged sports — boxing, gymnastics, diving — with much smaller or absent effects in objectively measured sports such as track and weightlifting. This decomposition provides an implicit test of the judging-bias mechanism that does not require access to individual judge scores.

Japan's National Sports Festival (国民体育大会, *Kokutai*; renamed 国民スポーツ大会 in 2024) offers an unusually clean setting for a Balmer-2003-style test. Since 1946, the Kokutai has rotated across all 47 prefectures on a fixed schedule, with essentially identical eligibility rules and a stable overall-score system. The event is large (approximately 40 sports across a summer meet and a smaller winter meet), long-running (80 editions through 2026), and centrally administered by the Japan Sport Association (JSPO). Two headline findings are widely known within Japan. First, host prefectures win at a startling rate: from 1978 through 2015, the host prefecture took the championship trophy in **97.3%** of editions (36 of 37 non-cancelled meets). Second, in a distinct pattern beginning in 2016, the host championship rate has collapsed to **37.5%** for 2016-2025 (3 of 8) — a discontinuity that has generated both journalistic speculation and one scholarly critique (Suetsugu, 2024, 2025)^{4,5} that host outcomes reflect “cheating” and a “host-victory-first mindset”. (For the historical record, the popular phrase “*Nara hantei*” — a slur for perceived hometown scoring bias — refers to a 2018 boxing scandal centered on Yamane Akira and the amateur boxing federation, not to the Kokutai kendo competition as is sometimes claimed.)

The most substantial quantitative predecessor is Funahashi, Hibino, Ishiguro & Mano (2016).⁶ They assembled a balanced 47-prefecture $\times$ 9-year panel ($n = 423$) for 2003-2011, regressed the male-female combined competition score on a full set of host-year event-time dummies ($t-7$ through $t+7$), and estimated a host-year bonus of +1,674.65 total-score points on top of prefecture and year fixed effects ($R^2 = 0.86$). Funahashi's Table 4 lists the six candidate mechanisms invoked above and cites Balmer et al. (2001)¹ — Balmer's Winter Olympics paper — as their reference for the subjective-judging pathway. But Funahashi et al. did **not** cite Balmer2003, and did not include either a sport-type indicator or a host $\times$ sport-type interaction in any of their specifications. The subjective-versus-objective decomposition that

Balmer2003 formalized is therefore absent from the Japanese panel literature. Chiba’s earlier qualitative treatment (1987)⁷ likewise enumerates five host-victory mechanisms — the full-entry system, alleged combination-drawing bias, transferred-athlete contributions, intra-prefecture athlete development, and prefecture-wide civic support — and explicitly does not mention refereeing or judging bias in any of the five.

Institutionally, the Japanese Kokutai stands in mild contrast to the Korean equivalent. The Korean Sport & Olympic Committee’s national comprehensive sports competition regulations explicitly award a **20% bonus** to the host city’s overall score, codified from 2010 onward (previously 10% from 2001).⁸ Japan has no such explicit host bonus in its scoring rules — the JSPO regulations we examined contain no analogous provision — which makes the Japanese host advantage, if present, a purely emergent phenomenon rather than an institutional artifact. This makes the Balmer2003-style objective/subjective decomposition especially informative in the Japanese case: an interaction that survives in the absence of an institutional bonus is more strongly suggestive of the crowd-and-judging pathway.

We accordingly ask three questions. First (**replication**), does a Kokutai host bonus of a similar order of magnitude survive an extension of Funahashi’s 2003-2011 panel through 2022, and does our reproduction of Funahashi’s specification recover his headline coefficient? Second (**structural change**), how should the discontinuity in host-victory rate around 2016 be characterized — as a single break, or as two clusters (2016-17 and 2022-24) separated by a return to host wins in 2018-19? Third and centrally (**novel test**), is the Kokutai host advantage larger in subjectively judged sports than in objectively measured ones, as Balmer2003 would predict? Our answers, in brief, are (i) yes: the host coefficient replicates within ~6% of Funahashi’s estimate; (ii) two clusters, not one, requiring a two-layer event-study design; and (iii) yes in direction and magnitude: the host $\times$ subjective interaction is +16.68 points on a base host effect of +16.60 (cluster-robust $p = 0.031$; wild-cluster bootstrap $p = 0.155$ under the few-treated-clusters correction), roughly doubling the host bonus in subjective sports relative to objective ones — with the descriptive host-boost gap (+37.9 subjective vs. +17.6 objective) monotonically ordered as Balmer2003 predicts, though the statistical significance of the point estimate should be read as tentative under the two-treated-prefecture design.

Methods

Data sources

The main analysis panel was assembled from three complementary sources. Overall prefecture rankings for the emperor’s cup (天皇杯; male-female combined) and empress’s cup (皇后杯; female only) from the 3rd through 79th editions were parsed from the Nagano Prefecture Sports Association ranked-listings page (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html), which reports the top 8 finishers per edition per trophy. Where an edition-specific total-score PDF was available on the JSPO official archive (<https://www.japan-sports.or.jp/kokutai/tabid183.html>), the individual-edition PDF was parsed with `pdfplumber` (Python 3.14) to recover per-prefecture total scores; the machine-readable PDFs covered

editions 58-67 (2003-2012) reliably, while editions 68-77 (2013-2022) were either 404 responses or image-only PDFs from which text could not be extracted (see Limitations). For the 2024 Saga (78th) and 2025 Shiga (79th) editions, JSPO published Excel files with the full 47-prefecture $\times$ 37-sport breakdown (78_score_data.xls, 78_score_all_data.xls, 79_score_all_data.xls, 79_score_data.xls); these were parsed with python-calamine (the Rust-based xlrd replacement, required because both files use a formula function ID that xlrd misidentifies).

Sport classification followed Balmer, Nevill & Williams (2003).³ Of the 40-41 Kokutai sports (37 summer + 3 winter, plus clay-target shooting in the 2024 Saga edition only), **17** were classified as **objective** (client-independent measurement: track and field, swimming, cycling, rowing, canoe, weightlifting, sailing, ski, skating, archery, kyudo, rifle shooting, clay-target shooting, golf, bowling, triathlon, sport climbing; clay-target shooting was appended in the 2024 edition and removed in 2025, so the effective objective count is 17 in 2024 and 16 in 2025), 11 as **subjective** (referee/judge scoring: kendo, judo, gymnastics, karate, jukendo, naginata, boxing, fencing, wrestling, sumo, equestrian), and 13 as **semi-subjective / team** (team ball sports with referee-mediated but primarily objective outcomes: soccer, basketball, volleyball, handball, hockey, rugby, softball, baseball, badminton, table tennis, soft tennis, tennis, ice hockey). The classification is implemented in src/sport_classifier.py and unit-tested against the Balmer2003 decision rules.

Two socioeconomic covariates were assembled from the Cabinet Office’s Economic and Social Research Institute (ESRI) prefectural accounts (令和4年度版; https://www.esri.cao.go.jp/jp/sna/data/data_list/kenmin/files/contents/main_2022.html): total population (soukatu9) and nominal prefectural GDP (soukatu1), both on the 2008SNA basis covering fiscal 2011-2022 for all 47 prefectures. Log-transformed versions were merged onto the panel via merge_confounders(); coverage is 100% (1,128 of 1,128) for non-special-edition cells in 2011-2022 and 100% NaN outside that window, restricting the covariate-adjusted analyses to 2012-2022.

Study design

The primary analysis is organized around three complementary specifications.

1. Main panel (2012-2022, n = 423 prefecture-years): Pooled ordered logit on rank (1-8, with “outside top 8” coded as rank 9) and pooled binary logit on top-1 (championship) status. Both pooled specifications include log-population and log-GDP as covariates by construction; the label “pooled” denotes the absence of prefecture and year fixed effects (FE), not the absence of covariates. The two primary pooled coefficients reported in Table 2 are therefore numerically identical to the corresponding M3 (+ log GDP) rows of the Csurilla-style staged specification (Table 4). The corresponding prefecture-and-year fixed-effect versions were fitted for comparability with Funahashi’s specification, but with fixed effects the small-sample maximum likelihood exhibited the expected complete-separation instability on top-1 (host prefectures win their host year almost deterministically, and Tokyo wins the majority of non-host years); we report the pooled specifications as primary and the FE specifications as sensitivity checks (with the FE specifications’ inflated point estimates reported honestly and interpreted as fingerprints of separation

rather than as substantively estimable effects). Standard errors for the pooled specifications are prefecture-clustered (47 clusters, Liang-Zeger sandwich estimator) to account for repeated observations of each prefecture across the 9 years of the panel; the fixed-effect specifications retain analytical (Fisher-information) SEs because prefecture FE renders further prefecture-clustering mechanically redundant, and their degenerate SEs are flagged as non-inferential regardless. The proportional-odds assumption underlying the ordered logit was probed with a Brant-style partial-PO diagnostic (each rank threshold fit as a separate prefecture-clustered binary logit, with the `is_host` coefficient compared across thresholds via an inverse-variance-weighted chi-square); the diagnostic and its tractability limitations on this dataset are reported in Limitations.

2. Funahashi (2016) replication (2003-2011, n = 423): Ordinary least squares (OLS) on total emperor's-cup score with prefecture and year fixed effects and a single host-year dummy — Funahashi's Table 4 “base” specification, but omitting Funahashi's population/GDP/headquarters/transfer-athlete/participants controls because ESRI 2008SNA GDP series do not extend before 2011. A pooled OLS without fixed effects and a +1-year-extended specification (2003-2012) are reported as sensitivity checks. The prefecture-clustered standard errors used for statistical inference are computed on 47 clusters. The purpose is validation of the pipeline against a published estimate, not a novel estimate.

3. 2024-2025 cross-section for the subjective-objective interaction (n = 6,991): Because sport-by-prefecture score data are only publicly released for the 2024 and 2025 editions, the Balmer2003-style host $\times$ sport-type interaction is estimated on a stacked cross-section. The theoretical panel size (a full cartesian product of prefectures and sport-year-trophy cells) is 47 prefectures $\times$ ([40 tennou + 35 kougou] in 2024 Saga + [40 tennou + 36 kougou] in 2025 Shiga) = 47 $\times$ 151 = **7,097 cells**; the regression frame retains **6,991 cells** (a 106-cell, 1.5% shortfall) after handling of the clay-target-shooting/boxing swap between the two editions and the drop of unbalanced cells for prefectures that did not participate in specific sports and are consequently absent from the JSPO overall-standings publication for that sport-year. The specification is $\text{score_isc} = \alpha + \beta_H \cdot \text{host_isc} + \beta_S \cdot \text{subj_c} + \beta_{HS} \cdot (\text{host} \times \text{subj})_isc + \kappa_c + v_t + \epsilon_isc$ where κ_c is a sport fixed effect and v_t is a year fixed effect. Prefecture-clustered robust standard errors are used (47 clusters). The primary specification uses a subjective binary indicator (subjective vs. non-subjective); a three-way interaction fully separating objective, semi-subjective, and subjective was fitted but produced a rank-deficient design matrix, so we report the three-way specification as descriptive only. A log-outcome specification ($\log(1 + \text{score})$) is reported as a robustness check.

Csurilla-style staged confounder controls

We fitted the sequence M1 (host only) $\rightarrow$ M2 (+ log population) $\rightarrow$ M3 (+ log GDP) $\rightarrow$ M4 (+ prefecture FE) $\rightarrow$ M5 (+ year FE), separately for ordered rank and top-1, on the 2012-2022 panel. Csurilla & Ferto (2023)⁹ report that adding population and per-capita GDP to a host-country regression attenuates the Olympic host coefficient by approximately 45% (from ~0.47 to ~0.26 for total medals). Whether the same

pattern holds at the Kokutai — where within-country socioeconomic variance is smaller than within-Olympic-country variance — is not obvious *ex ante*.

Event-study design (two-layer)

An initial single-break difference-in-differences (DID) centered at 2016 was rejected because the raw data show host wins in both 2018 (Fukui) and 2019 (Ibaraki), i.e., a return to the pre-2016 pattern within the “post” period — this violates the parallel-trend assumption of a single-break design. We instead specified a two-layer event-study following the Phase 8 investigation’s H-03 recommendation: **Layer 1** (pre-2005 regime) contains the single 2002 Kochi host-loss event as an isolated pre-*furusato*-athlete-rule (introduced 2005) shock, and **Layer 2** (post-2016 regime) stacks the five host-loss events of 2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima (delayed special edition), and 2024 Saga. The event-study is fitted as a linear-probability model (top-1 outcome) with prefecture-clustered SEs; formal parallel-trend Wald tests are reported. As will be evident in the Results section, the identifying variation in these event-study cells is thin because the treated group (host prefectures) has non-zero top-1 status essentially only in the host year itself, so the event-study estimate is best interpreted as a descriptive complement to the main-panel result rather than as an independent identifying analysis.

Statistical software

All analyses were performed with Python 3.14 using pandas 3.0.5, numpy 2.5.1, statsmodels 0.14.6, patsy 1.0.2, scipy 1.18.0, python-calamine 0.8.2, pdfplumber 0.11.10, and matplotlib 3.11.1. Prefecture-cluster-robust standard errors (47 clusters, Liang-Zeger sandwich) are reported for all main-panel pooled specifications, the Csurilla-style M1-M3 stages, the replication OLS, and the 2024-2025 cross-section; Fisher-information standard errors are retained for the M4/M5 fixed-effect specifications where prefecture FE renders further clustering mechanically redundant and where complete separation drives SEs to numerical degeneracy in any case. Statistical tests are two-sided at $\alpha = 0.05$. Unit tests ($n = 281$ total, all passing) verify the numerical integrity of each analysis module against small deterministic fixtures.

Multiple comparisons. The staged analytic plan treats the 2024-2025 cross-section host $\times$ subjective interaction (β_{HS} , Table 5 baseline row) as the single confirmatory test of the paper’s central hypothesis (that the Kokutai host bonus is larger in subjectively judged sports than in objectively measured ones). The pooled main-panel host coefficients (Tables 2 and 4), the Funahashi replication (Table 3), the Csurilla-style staged confounder specifications (Table 4 and Table 4b), and the two-layer event-study (Figure 3) are treated as replication, sensitivity, and design-check specifications respectively — they are not independent confirmatory tests of the central hypothesis and do not each require their own α -level adjustment. Consequently, we do not apply Bonferroni or Benjamini-Hochberg correction across the roughly 15 reported p-values in the manuscript. The confirmatory hypothesis is tested exactly once, and its outcome under the appropriate few-treated-clusters correction (wild-cluster bootstrap $p = 0.155$; see Finding #5-linked paragraphs in Results and Limitations) is reported directly. A reader preferring an explicit Bonferroni-style guardrail may note that even a very generous 5-test correction ($\alpha_{\text{family}} = 0.05 \rightarrow \alpha_{\text{single}} = 0.01$) would not change the qualitative reading: the naive cluster-robust p (0.031) would

remain below the corrected 0.01 threshold only marginally, and the bootstrap p (0.155) would remain well above it under either correction.

Ethical considerations

This study analyzes publicly available Kokutai overall-standings data released by the Japan Sport Association and prefectural sport associations. No individual-level, health, or personally identifying data are used, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public aggregate competition results without any personally identifiable information, it does not meet the standard threshold for human-subjects research under either domestic (e.g., MEXT/MHLW ethical guidelines for research using publicly available secondary data) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required.

Results

Descriptive statistics (2012-2022 panel)

Across 2012-2022, the emperor's-cup panel contains 423 prefecture-year observations (47 prefectures $\times$ 9 non-special, non-cancelled years). Host prefectures ($n = 9$ host-years) took the championship in 6 of 9 host-years (66.7%) — a rate two orders of magnitude above the non-host per-year top-1 rate of 3 in 414 (0.72%) — and appeared in the top 8 in **9 of 9 host-years (100%)**. This 100% top-8 rate constitutes a complete separation in a top-8 logit and was accordingly handled descriptively rather than as a fitted model: the fact of nine-for-nine top-8 host appearances is itself a stronger point estimate than any logit could produce on such a fixture. When the host prefecture did finish inside the top 8 (which is to say, always), the mean host rank was 1.33.

The three host-loss host-years within the 2012-2022 panel window (in which a non-host prefecture — in each case Tokyo — took the championship) were 2016 Iwate, 2017 Ehime, and 2022 Tochigi. Kagoshima's 2023 special edition (a COVID-delayed re-run of the cancelled 2020 program) and 2024 Saga also produced Tokyo championships outside the panel window, extending the run to 2022-24 in the descriptive time series. The six host-victory host-years within the panel window fall in 2012 Gifu, 2013 Tokyo (self-host), 2014 Nagasaki, 2015 Wakayama, 2018 Fukui, and 2019 Ibaraki (six wins in the six non-loss host-years within the panel window; the 2013 Tokyo cell is Tokyo hosting itself and is retained in the regression sample but is not used as a host-effect data point in the “6 of 9” descriptive rate because Tokyo dominates non-host years generally).

Main panel: pooled ordered logit and top-1 logit (with log-population and log-GDP controls)

The pooled ordered logit on rank (censored at 9 = outside top 8), with log-population and log-GDP controls, yielded a host coefficient of **-13.73** (SE = 2.03, prefecture-clustered; $z = -6.76$; $p < 0.001$; $n = 423$), where a negative coefficient corresponds to better (numerically smaller) rank on the ordered scale.

In log-odds terms this is an extraordinarily large shift — the median host prefecture-year sits at rank 1, versus a median non-host prefecture-year outside the top 8 — but the magnitude reflects the near-deterministic character of host top-8 status in the data rather than a subtle behavioral effect. The pooled binary logit on top-1 (same controls) gave a host coefficient of **+9.09** (SE = 2.13, prefecture-clustered; $z = 4.28$; $\mathbf{p} < \mathbf{0.001}$, exact $p = 1.9 \times 10^{-5}$), corresponding to an odds ratio far above 100 that is best interpreted, again, as a fingerprint of the concentrated top-1 mass in the host-year cells rather than a stable behavioral log-odds. These two pooled coefficients are numerically identical to the M3 (+ log GDP) row of Table 4 and share the same prefecture-clustered SE; the “pooled” label denotes the absence of prefecture and year fixed effects only, not the absence of covariates or of clustering (see Methods §1). Under the naive Fisher-information SE that would ignore the repeated-observations structure of each prefecture across years, the corresponding SEs would be 1.54 for ordered rank and 2.80 for top-1 — the ordered-rank SE understates prefecture-level dependence while the top-1 SE overstates it, but neither correction changes the direction or the significance of the effect at $\alpha = 0.05$.

The prefecture-and-year fixed-effect versions, which more closely mirror Funahashi’s specification, produced degenerate estimates as expected under complete or near-complete separation: the FE ordered logit gave a host coefficient of -49.56 (SE = 172.45; $p = 0.77$), a fingerprint of the optimizer’s failure to identify a meaningful ordered-logit slope when the treated-versus-untreated cells within each prefecture and year are near-singleton; the FE top-1 logit gave a coefficient of +1,877 (SE = NaN), a classic separation blowup. These FE specifications are reported for transparency but should not be interpreted as coefficient estimates in the ordinary sense.

Replication of Funahashi et al. (2016), 2003-2011

Our reproduction of Funahashi’s prefecture-and-year fixed-effect OLS on emperor’s-cup total score for 2003-2011 ($n = 423$) yielded a host coefficient of **+1,575.45** total-score points (SE = 57.58, based on 47 prefecture clusters; $p < 0.001$; $R^2 = 0.94$). Funahashi’s Table 4 reports a corresponding base-specification coefficient of +1,674.65, so our reproduction lies within 5.9% of Funahashi’s estimate. The residual gap is attributable to our omission of Funahashi’s socioeconomic controls (population, GDP, headquarters count, transfer-athlete rule ratio, participant count), which we could not include because ESRI 2008SNA prefectural GDP series do not extend before fiscal 2011. Sensitivity checks corroborated the finding: a pooled OLS without fixed effects gave a coefficient of +1,733.29 (SE = 138.26), and a +1-year extension to 2003-2012 ($n = 470$) gave +1,604.36 (SE = 60.08). Together, the three specifications bracket Funahashi’s headline value tightly. Figure 5 displays the three replication coefficients alongside Funahashi’s +1,674.65 reference line, with 95% confidence intervals ($1.96 \times \text{SE}$) shown as error bars. The `pooled_no_fe` specification lies within one SE of the reference, and the `funahashi_base` and `extended_2003_2012` specifications lie within 1.96 SE (95% CI) of the reference.

Csurilla-style staged confounder controls

The staged specification produced a *strengthening* rather than an attenuation of the host coefficient as socioeconomic controls were added, running in the opposite direction from the ~45% attenuation Csurilla

& Fertő (2023) documented for Olympic host countries. This directional comparison is not a scale-comparable comparison: Csurilla & Fertő fit a zero-inflated negative-binomial model on total medal counts (log link, country-by-Olympics identifying variance), whereas our specifications are a binary logit on top-1 status and an ordered logit on rank (logit link, prefecture-by-Kokutai identifying variance). The absolute-percentage-change directions differ, but the two coefficients live on different scales and identify off different variance decompositions, so the comparison is best read as “the *direction* of the socioeconomic-controls effect on the host coefficient is inverted, on this paper’s own scale” rather than as an IRR-to-IRR opposition. For the top-1 outcome, the M1 (host-only) coefficient was +5.61 (SE = 1.09, prefecture-clustered; $p < 0.001$); adding log population (M2) raised the coefficient to +10.90 (SE = 3.35; $p = 0.0011$); adding log GDP (M3) settled it at +9.09 (SE = 2.13; $p < 0.001$, exact $p = 1.9 \times 10^{-5}$). M4 (prefecture FE) and M5 (year FE) again produced separation-driven blowups (+498 and +1,877 respectively, with degenerate SEs; Fisher-information reported because prefecture FE renders clustering mechanically redundant) that are not interpretable as attenuation. For the ordered-rank outcome the same directional pattern held: M1 = -6.11 (SE = 1.21), M2 = -13.17 (SE = 2.01), M3 = -13.73 (SE = 2.03), all $p < 0.001$; a $2.25\times$ *increase* in absolute magnitude from M1 to M3. Figure 4 displays the M1→M3 trail for the top-1 outcome, showing an anti-attenuation trajectory rather than the Csurilla-style ~45% decay. The most natural interpretation is that population and GDP are *negatively* correlated with the host coefficient’s identifying variation at the Kokutai — Tokyo, an outlier on both covariates, is a large-population high-GDP prefecture that dominates non-host years, so controlling for population and GDP absorbs some of the non-host championship mass and (in relative terms) makes the host coefficient look larger. This is the opposite fingerprint from an Olympic setting in which the host is typically the largest and richest country in a given year: at the Kokutai, host prefectures are essentially random draws from the size distribution, so socioeconomic controls do not attenuate the host coefficient.

Event-study, two-layer design

The two-layer event-study — Layer 1 (2002 Kochi single event) and Layer 2 (2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga) — was fitted as a linear-probability model on top-1 status. The parallel-trend Wald tests were formally non-rejecting in both layers (Wald $p = 1.0$), but the identifying variation is thin: the treated group (host prefectures) exhibits non-zero top-1 status essentially only in the host year itself, and the pre- and post-shock cells for the treated group are mechanically zero. As a result, the event-study point estimates are near-zero by construction and should be read as a design check rather than as an independent identifying analysis of the shock timing. Figure 3 displays the τ -coefficient trails for both layers alongside their 95% confidence bands. This limitation is a fundamental consequence of the top-1 outcome variable being effectively a host-year indicator; a continuous-score outcome would in principle allow a stronger event-study, but the 2003-2011 replication window is where continuous-score data are available, and that window contains no post-2016 shocks.

2024-2025 cross-section: the subjective-objective interaction

The stacked 2024-2025 cross-section ($n = 6,991$ prefecture-sport-year-trophy cells; 47 prefectures $\times$ 40 - 41 sports $\times$ 2 years $\times$ 2 trophies, after handling of the clay-target/boxing swap between the two editions) is the specification in which the Balmer2003-style objective/subjective decomposition can be estimated. The baseline specification, controlling for sport and year fixed effects with prefecture-clustered SEs, produced a host main effect of **+16.60 points per sport** ($SE = 1.25$; $p < 0.001$) and a **host $\times$ subjective interaction of +16.68 points** ($SE = 7.72$; cluster-robust $p = 0.031$). Because the main effect is a within-sport, cross-prefecture difference, +16.60 points is roughly the average per-sport score bonus a host prefecture receives in a non-subjective sport; +16.68 is the *additional* bonus in a subjective sport. Together, the model implies a total per-sport host bonus of roughly 33 points in a subjectively judged sport versus about 17 in an objectively measured one — essentially double.

The 2-of-47 treated-cluster structure of this cross-section (only Saga in 2024 and Shiga in 2025 are host prefectures) exposes the cluster-robust variance formula to the Cameron & Miller (2015)¹³ few-treated-clusters problem: with so few treated clusters, the cluster-robust standard error tends to be anti-conservative and the associated t-statistic is not well-approximated by its asymptotic distribution. We accordingly computed a wild-cluster bootstrap p-value (Cameron, Gelbach & Miller, 2008)¹⁴ using Rademacher weights on prefecture clusters, a restricted null ($\beta_{HS} = 0$), and $B = 999$ iterations. The bootstrap p for the interaction was $p = 0.155$ (observed $t = 2.16$), which does *not* attain conventional (0.05) significance under this correction. The direction and magnitude of the interaction are preserved, but the point estimate should be read as a first empirical anchor at this stage of the data availability rather than as a decisive rejection of the null; the descriptive +37.9 vs. +17.6 subjective/objective gap and the log-outcome interaction coefficient (+0.293, same direction, cluster-robust $p = 0.084$) both support the Balmer2003 direction independently of the p-value scale.

The three-way specification with a semi-subjective interaction was rank-deficient (multicollinearity between the sport-fixed-effects and the three-way subjective/semi-subjective/objective decomposition) and is not reported as inferential; the log-outcome specification returned a host coefficient of +0.467 ($SE = 0.057$; $p < 0.001$) and an interaction coefficient of +0.293 ($SE = 0.170$; $p = 0.084$, marginal), directionally consistent with the linear specification.

Descriptively, the raw host-versus-nonhost per-sport score gap monotonically increased across the three sport categories in exactly the direction Balmer2003 predicts (Figure 2): objective sports showed a gap of **+17.60 points** (host mean 43.63 vs. non-host mean 26.03, $n_{\text{host}} = 65$ cells), semi-subjective/team sports **+26.18 points** (47.03 vs. 20.85, $n_{\text{host}} = 48$), and subjective sports **+37.91 points** (57.24 vs. 19.33, $n_{\text{host}} = 38$). The subjective host boost of +37.9 is the largest of the three, and its gap over the objective +17.6 (a ratio of 2.15) is close to the model-implied doubling. Figure 2 displays the three category-specific host boosts with 95% confidence intervals.

Structural change: two clusters, not one

The raw host-championship rate declines from 97.3% for 1978-2015 (36 of 37 non-cancelled meets) to 37.5% for 2016-2025 (3 of 8 non-cancelled meets). This is consistent with a structural break around 2016. However, decomposing the 2016-2025 window reveals *two* clusters of host losses (2016-17 and 2022-24) separated by a full return to host wins in 2018 Fukui (host score 2,896 vs. Tokyo's 2,246) and 2019 Ibaraki (2,569 vs. 2,217). A single-break DID would blur these two clusters into one and misspecify the intermediate 2018-19 period. The two-layer event-study specification described in Methods was adopted precisely to preserve this structure. Figure 1 displays the emperor's-cup host championship-rate time series with the six host-loss years (2002 Kochi, 2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga) marked; the two post-2016 clusters and the 2018-19 island are clearly visible.

Discussion

The Balmer2003 decomposition holds at the Kokutai

The central finding of this study is that the Kokutai host bonus appears to be not uniform across sport types: the marginal bonus in subjectively judged sports is roughly double the bonus in objectively measured sports. In the 2024-2025 stacked cross-section, the host main effect is +16.60 points per sport and the host $\times$ subjective interaction adds a further +16.68 points (cluster-robust $p = 0.031$; wild-cluster bootstrap $p = 0.155$ under the few-treated-clusters correction), yielding a model-implied total per-sport bonus of roughly 33 points in subjective sports against ~ 17 in objective ones. The raw descriptive gap orders the three sport categories monotonically as +37.9 (subjective) $>$ +26.2 (semi-subjective) $>$ +17.6 (objective) — precisely the ordering that Balmer, Nevill & Williams (2003) documented at the summer Olympics. Given that the wild-cluster bootstrap does not reach conventional significance under the two-treated-cluster structure of this cross-section, we frame this as a *directionally* consistent first quantitative anchor for the Balmer2003 decomposition on Kokutai data — supported concordantly by (a) the raw descriptive monotonicity, (b) the log-outcome interaction direction, and (c) the linear interaction point estimate — rather than a decisive statistical confirmation. It nevertheless closes the specific gap left by Funahashi, Hibino, Ishiguro & Mano (2016), whose 2003-2011 panel documented a robust overall host bonus but did not disaggregate by sport type.

The finding is consistent with, but does not require, a judging-bias mechanism. The Balmer2003 decomposition is an *implicit* test — an interaction of the observed type is what a judging-bias mechanism would produce, but a purely crowd-effect mechanism that is stronger in enclosed, spectator-dense subjective-sport venues (typical: judo, kendo, gymnastics halls) than in open-air objective-sport venues (typical: track and field, cycling road courses) would produce the same pattern. Without a judge-level dataset we cannot uniquely identify judging bias as the mechanism. What we can say is that a purely material mechanism — training-venue familiarity, budget, transfer-athlete rules — should not produce a differential effect across sport types, and the interaction we recover is more consistent with a crowd-and-judging pathway than with a pure material one. Nomura (2022)¹⁰ provides collateral evidence from

Japanese J1 football that home advantage disappeared during the reduced-occupancy 2020 COVID-19 season and — through multiple-group structural equation modeling — that the crowd’s influence on referees’ decisions vanished under the same conditions, further supporting the crowd–referee pathway.

Two host-loss clusters, not one

The popular framing of a “post-2016 Tokyo winning streak” is factually incorrect. Host prefectures won both 2018 Fukui (2,896 vs. Tokyo’s 2,246) and 2019 Ibaraki (2,569 vs. Tokyo’s 2,217) under exactly the same national scoring rules that produced the Tokyo wins in 2016 Iwate and 2017 Ehime. The 2020 and 2021 editions were cancelled for COVID-19, with the 2020 Kagoshima program re-scheduled as a special edition in 2023. The correct structural pattern is therefore two clusters of host losses (2016-17 and 2022-24) separated by a return to normal pattern, plus the single 2002 Kochi outlier in the pre-2005 regime. The two-layer event-study specification we adopt preserves this structure; a naive single-break DID at 2016 would produce a biased estimate of the structural change. This is a modeling recommendation for future Japanese work on the Kokutai host effect: do not treat 2016 as a single break.

East-Asian institutional context

The Kokutai host bonus emerges *in the absence of* an explicit institutional host bonus in JSPO scoring rules. This stands in contrast to the Korean Sport & Olympic Committee’s national comprehensive sports competition, which explicitly adds a 20% score bonus to the host city (10% from 2001, raised to 20% in 2010),⁸ and to the qualitative characterization Lan and Yu (2012)¹¹ provide of the Chinese National Games host effect, whose five listed characteristics — high efficacy, gradual increase, delayed effect, exclusivity, and double-edgedness — describe roughly the same phenomenon we observe empirically at the Kokutai but in more diffuse qualitative language. The present study is arguably an implicit decomposition of Lan & Yu’s “exclusivity” and “double-edgedness” characteristics: exclusivity because the host bonus is concentrated in a subset of sports (subjectively judged), and double-edgedness because a large host bonus in a subjective sport increases both the host’s expected medal count and the perceived legitimacy risk of that medal (see the Suetsugu 2024/2025 qualitative critique below).

Overlap and boundary with Suetsugu (2024, 2025)

The Suetsugu papers^{4,5} provide a karate-specific qualitative critique of Kokutai outcomes, using pointed language (“cheating” / “host-victory-first mindset”) in a single-sport context. Our whole-sport panel-quantitative decomposition is complementary rather than overlapping: Suetsugu’s papers focus on a single subjectively judged sport (karate) and characterize the outcome pattern in that sport in normative terms; our analysis provides the sport-level statistical evidence that the pattern Suetsugu describes in karate is indeed sport-type-specific (larger in subjectively judged sports than in objectively measured ones) rather than merely idiosyncratic to karate. The two contributions can stand side by side: Suetsugu establishes the qualitative concern within a single sport; we establish the quantitative interaction across all sports.

We deliberately avoid the normative language Suetsugu uses. The observed interaction is consistent with judging bias, but it is also consistent with subject-sport-specific crowd effects, subject-sport-specific practice-environment effects, and subject-sport-specific transfer-athlete effects. The present study does not adjudicate among these mechanisms; it establishes only that they are collectively larger in subjectively judged sports than in objectively measured ones, by a factor of approximately two.

Robustness to socioeconomic controls

Csurilla & Fertő (2023)⁹ report that adding population and per-capita GDP controls to a country-level Olympic host regression attenuates the host coefficient by approximately 45%. Our Kokutai analog shows the *inverse*: adding population and GDP to the M1 host-only specification *increases* the absolute magnitude of the host coefficient by roughly a factor of 2.25 (for the ordered-rank outcome, from -6.11 to -13.73). This inversion is neither an error nor a paradox: it reflects that at the Kokutai the host is essentially a random draw from the size distribution of prefectures — the rotation schedule is fixed decades in advance and does not correlate with prefectural size — whereas Tokyo, an outlier on both population and GDP, dominates the non-host championship mass. Controlling for size therefore absorbs part of the non-host championship variance and (in relative terms) makes the host effect look larger. This mechanism is quantitatively confirmed by a Tokyo-exclusion sensitivity check (Table 4b): removing Tokyo from the panel ($n = 414$ vs. 423) purges all three non-host top-1 wins (2016, 2017, 2022 — all Tokyo) and produces complete separation on top-1, roughly doubling the M3 host coefficient (top-1 $+9.09 \rightarrow +17.66$; ordered rank $-13.73 \rightarrow -24.00$). The doubled coefficient should be read as directional evidence of the Tokyo-driven mechanism rather than as a preferred point estimate, since the separation regime distorts both the Fisher-information SE and the point estimate. This is a substantive finding: the Kokutai host bonus is *not* driven by prefectural size or economic capacity, and remains robust to standard socioeconomic controls.

Limitations

Eight limitations of the present study should be noted.

First, edition-level sport-by-prefecture score data are only publicly available for the 2024 Saga and 2025 Shiga editions. All earlier editions (through 2023 Kagoshima special) publish only prefecture-level overall standings, not the sport-by-prefecture breakdown. This restricts the Balmer2003-style objective/subjective interaction test to a two-year cross-section ($n = 6,991$) with only **2 treated prefecture clusters (Saga, Shiga) out of 47**. Under Cameron & Miller’s (2015)¹² few-treated-clusters diagnosis, the cluster-robust variance formula is expected to be anti-conservative in this regime, and we explicitly report the wild-cluster bootstrap p-value ($p = 0.155$) alongside the cluster-robust p ($p = 0.031$) in the Results section; the interaction magnitude should therefore be viewed as a first empirical anchor rather than a definitive magnitude. Year fixed effects in the cross-section are effectively a single dummy given the two-year window. If JSPO releases historical sport-by-prefecture breakdowns in the future, extending the

interaction to a longer panel would substantially increase the treated-cluster count and tighten both the estimate and its inferential support.

Second, we could not obtain the JSPO judge-of-record roster (name + prefecture of affiliation) for any Kokutai edition. Four independent search paths — JSPO regulations catalog `tabid188`, seven individual-edition sampling, kendo-specific archives, and general web search — all returned negative. The Balmer2003-style decomposition is therefore the only implicit test of judging bias we can offer; a direct judge-level Zitzewitz (2006)-style analysis¹² is not available for Kokutai data at present.

Third, the subjectively judged sports for which we could confirm high-fidelity edition-level score data (independent of the JSPO overall-standings pipeline) are essentially limited to kendo, which published a per-prefecture score table for the 2016 Iwate edition (71st) but discontinued the same table by 2022 Tochigi (77th). Judo, gymnastics, and other subjectively judged sports were not covered in this way. The 2024-2025 cross-section provides sport-level scores, but the pre-2024 subjective-sport history is patchy.

Fourth, the Suetsugu (2024, 2025) full texts are served through the Komazawa University institutional repository as WEKO3 octet-stream downloads that could not be reliably extracted; the citations are at bibliographic-plus-CiNii-link level only. This is a citation-quality limitation, not a data limitation, and does not affect the present study's numerical results.

Fifth, the Funahashi (2016) replication reported here omits Funahashi's population, GDP, headquarters, transfer-athlete, and participant controls because the ESRI 2008SNA prefectural GDP series (the reference socioeconomic dataset for our main-panel analysis) does not extend to fiscal 2002 or earlier. Our replicated coefficient of +1,575.45 accordingly represents a simpler specification than Funahashi's; the 5.9% gap versus Funahashi's +1,674.65 is well within what would be expected from the omitted controls, but a formal all-controls-included replication would require assembly of pre-2011 prefectural socioeconomic covariates from a separate ESRI historical release, which was outside the scope of the present study.

Sixth, the primary analysis panel window is 2012-2022 (11 years, $n = 423$), constrained by the ESRI 令和4年度版 (2011-2022) coverage that is the newest release available as of manuscript preparation. When ESRI releases the 令和5年度版 (extending through fiscal 2023), the panel can be extended by one year with essentially the same code path; when the 令和6年度版 releases (extending through 2024), the panel can be extended to include the 2024 Saga edition (currently reported only in the 2024-2025 cross-section). Extending the panel to 2024 would allow the 2024 Saga host-loss event to enter the main-panel analysis directly, tightening the event-study specification's identifying variation on the post-2016 cluster.

Seventh, we ran a Brant-style partial-proportional-odds diagnostic on the pooled ordered logit by fitting a separate binary logit at each rank threshold ($Y \leq j$ vs. $Y > j$, for $j = 1, \dots, 8$) with prefecture-clustered standard errors and comparing the `is_host` coefficient across thresholds via an inverse-variance-weighted chi-square. Only the top-1 threshold ($Y \leq 1$) yielded a converged fit ($\beta = +9.09$, $SE = 2.13$, matching the top-1 pooled logit exactly); all higher thresholds ($Y \leq 2$ through $Y \leq 8$) hit numerical singularity because every host prefecture-year in the panel finishes in the top 8 (`host_top8_rate` = $9/9 = 100\%$), leaving no

within-cell variance for those threshold-specific binary logits. A formal test of the proportional-odds assumption is therefore not tractable on this data-generating regime. Qualitatively, the same near-complete separation is itself circumstantial evidence that a single-slope representation is not violated by threshold heterogeneity — the host effect is so concentrated at the top that there is essentially no room for β_j to vary from threshold to threshold — but we cannot formally certify the assumption. We report the diagnostic honestly and acknowledge the limitation; the substantive conclusions in this paper rest on the pooled top-1 logit (which does not depend on the PO assumption at all), the pooled ordered logit (whose robustness we cannot formally test but which produces coefficients consistent with the top-1 fit), the Csurilla-style staged specification, and the 2024-2025 cross-section, and are therefore not overturned by the untestable status of the PO assumption in this particular ordered-logit specification.

Eighth, the β_{HS} (host $\times$ subjective interaction) coefficient in the 2024-2025 cross-section rests on an asymmetric identification structure that is worth making fully explicit alongside the few-treated-clusters caveat already noted in the First limitation above. In practical terms, the interaction is identified off the difference between Saga and Shiga in their subjective-vs-objective host-cell contrasts: roughly 38 subjective host cells (Saga 2024 subjective sports + Shiga 2025 subjective sports) carry essentially the entirety of the interaction identification, with the roughly 65 objective host cells and the remaining semi-subjective cells serving as within-prefecture reference groups. This is not a generic “2 of 47” cross-section design; it is a specific 2-prefecture design in which the interaction’s variance is a function of exactly how those two prefectures happened to differ in subjective-sport versus objective-sport performance during their respective host years. If Saga and Shiga had turned out to be substantively similar in their subjective-sport performance (a plausible counterfactual given both are mid-sized prefectures without an obviously dominant elite-athlete pipeline in judo/kendo/gymnastics), the interaction would be estimated off a small residual signal and the true SE would be correspondingly inflated. This asymmetric identification is one of the mechanical reasons the wild-cluster bootstrap p (0.155) diverges so substantially from the naive cluster-robust p (0.031) — the bootstrap re-samples over prefecture clusters and honestly reflects the fragility of a two-cluster identification base, whereas the cluster-robust SE leans on asymptotic approximations known to be anti-conservative in this regime (Cameron & Miller 2015)¹³. Extending the interaction test to a longer panel with more host prefectures would replace the current pairwise Saga-vs-Shiga structure with a broader within-prefecture identifying variance and correspondingly relax both the anti-conservative-SE concern and the two-prefecture asymmetric-identification concern.

Conclusion

Japan’s National Sports Festival exhibits a host bonus that is directionally not uniform across sport types. The marginal host bonus in subjectively judged sports is roughly double the bonus in objectively measured sports (host $\times$ subjective interaction = +16.68 points on a host main effect of +16.60 points; cluster-robust p = 0.031; wild-cluster bootstrap p = 0.155 under the few-treated-clusters correction on 2 treated prefectures out of 47), matching the direction of the Balmer, Nevill & Williams (2003) decomposition established at the summer Olympics. The overall host bonus is robust to prefecture-level

population and GDP controls (indeed strengthened by them, unlike at the Olympics), replicates Funahashi et al. (2016)'s 2003-2011 headline coefficient within 5.9%, and survives an extension of Funahashi's panel through 2022. The structural change in host championship rate around 2016 is best characterized as two clusters (2016-17 and 2022-24) separated by a return to host wins in 2018-19, not a single break. The finding closes the specific quantitative gap left by Funahashi (2016) and provides Japanese panel-econometric evidence complementary to the qualitative critique of Suetsugu (2024, 2025). Whether the observed interaction reflects judging bias, subject-sport-specific crowd effects, or subject-sport-specific practice-environment effects cannot be uniquely identified from the present design; the interaction itself, however, is now empirically anchored.

References

1. Balmer NJ, Nevill AM, Williams AM. Home advantage in the Winter Olympics (1908-1998). *J Sports Sci.* 2001;19(2):129-139. doi:10.1080/026404101300036334
2. Wilson D, Ramchandani G. A comparative analysis of home advantage in the Olympic and Paralympic Games 1988-2018. *J Global Sport Manage.* 2021;6(2):170-184. doi:10.1080/24704067.2018.1537676
3. Balmer NJ, Nevill AM, Williams AM. Modelling home advantage in the Summer Olympic Games. *J Sports Sci.* 2003;21(6):469-478. doi:10.1080/0264041031000101890
4. Suetsugu M. [Issues emerging from the karate competition results of the National Sports Festival]. *Bulletin of the Institute of Liberal Arts and Sciences, Komazawa University.* 2024;18:137-153. CiNii: <https://cir.nii.ac.jp/crid/1390303254959134848> (Article in Japanese; not indexed on Crossref).
5. Suetsugu M. [Visualization and structural analysis of the issues in karate competition at the National Sports Festival]. *Bulletin of the Institute of Liberal Arts and Sciences, Komazawa University.* 2025;19:103-117. doi:10.69200/0002033886 (Article in Japanese; Komazawa institutional DOI, not indexed on Crossref).
6. Funahashi H, Hibino M, Ishiguro E, Mano Y. Determinants of success at the National Sports Festival. *Japanese Journal of Sport Management.* 2016;8(1):17-33. doi:10.5225/jjism.2016-002 (Article in Japanese; English title from J-STAGE record).
7. Chiba T. [Factors underlying host-prefecture victories at the National Athletic Meet]. *Proceedings of the Japanese Society of Physical Education.* 1987;38a:204. doi:10.20693/jspeconf.38a.0_204 (Japanese).
8. Kim T. [KSOC national comprehensive sports competition host-city 20% bonus regulation]. *Munhwa Ilbo.* Published 2025-10-13. <https://v.daum.net/v/20251013192536045> (Korean).
9. Csurilla G, Fertő I. The less obvious effect of hosting the Olympics on sporting performance. *Scientific Reports.* 2023;13:819. doi:10.1038/s41598-022-27259-8

10. Nomura R. Influence of crowd size on home advantage in the Japanese football league. *Frontiers in Sports and Active Living*. 2022;4:927774. doi:10.3389/fspor.2022.927774
 11. Lan T, Yu X. [Theoretical and empirical research on the characteristics of the host effect at the National Games of China]. *Journal of Shenyang Sport University*. 2012;31(4):1-5. Abstract retrieved from <https://m.fx361.com/news/2012/1109/12534183.html> (Article in Chinese; Chinese-local journal, not indexed on Crossref; primary text not accessible outside Chinese academic networks).
 12. Zitzewitz E. Nationalism in winter sports judging and its lessons for organizational decision making. *J Econ Manage Strategy*. 2006;15(1):67-99. doi:10.1111/j.1530-9134.2006.00092.x
 13. Cameron AC, Miller DL. A practitioner's guide to cluster-robust inference. *J Human Resources*. 2015;50(2):317-372. doi:10.3368/jhr.50.2.317
 14. Cameron AC, Gelbach JB, Miller DL. Bootstrap-based improvements for inference with clustered errors. *Rev Econ Stat*. 2008;90(3):414-427. doi:10.1162/rest.90.3.414
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Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

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Data Availability

All input data used in this study are publicly available. Prefecture rankings are from the Nagano Prefecture Sports Association (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html); individual-edition PDFs and 2024-2025 Excel breakdowns are from the JSPO official archive (<https://www.japan-sports.or.jp/kokutai/tabid183.html>); socioeconomic covariates are from the Cabinet Office ESRI prefectural accounts (https://www.esri.cao.go.jp/jp/sna/data/data_list/kenmin/files/contents/main_2022.html).

Analysis code (Python 3.14, pandas 3.0.5, statsmodels 0.14.6, matplotlib 3.11.1, weasyprint 69.0) — including the data-assembly scripts, all regression modules (pooled ordered/binary logit, Csurilla-style staged specification, Funahashi replication OLS, 2024-2025 cross-section), the wild-cluster bootstrap and Brant-style partial-proportional-odds diagnostic implementations, the plotting pipeline, and the PDF build pipeline — will be released to a public GitHub repository and permanently archived on Zenodo (with a DOI) concurrent with, or within 30 days of, SSRN posting. This commitment is made explicit to avoid an asymmetric-disclosure gap in which AI-usage transparency (see Acknowledgments — LLM-assisted development is fully disclosed at the point of first posting) would run ahead of code transparency and thereby impair independent reviewer verification of the LLM-assisted numerical results reported here.

Tables

Table 1. Descriptive statistics for the 2012-2022 emperor’s-cup panel

Statistic	Value
Number of prefecture-year observations	423
Number of host-year cells	9
Number of non-host prefecture-year cells	414
Host cells that finished in top 1 (championship)	6 (66.7%)
Host cells that finished in top 8	9 (100%)
Non-host cells that finished in top 1	3 (0.72%)
Non-host cells that finished in top 8	62 (15.0%)
Host mean rank when ranked (top 8)	1.33

The 9 host years in the panel window (2012-2022) are 2012 Gifu, 2013 Tokyo, 2014 Nagasaki, 2015 Wakayama, 2016 Iwate, 2017 Ehime, 2018 Fukui, 2019 Ibaraki, and 2022 Tochigi ($n = 9$ non-cancelled non-special host years; 2020 Kagoshima and 2021 Mie were cancelled for COVID-19, and 2023 Kagoshima special and 2024 Saga fall outside the 2012-2022 window). Of the 9 host years, 6 were host victories (2012 Gifu, 2013 Tokyo self-host, 2014 Nagasaki, 2015 Wakayama, 2018 Fukui, 2019 Ibaraki)

and 3 were host losses (2016 Iwate, 2017 Ehime, 2022 Tochigi — all won by Tokyo). The 2013 Tokyo cell is retained in the regression sample ($n = 423$) but is not used as a host-effect data point in the “6 of 9” descriptive rate reported in Results because Tokyo dominates non-host years generally.

Table 2. Main-panel regression results (2012-2022 tennou, $n = 423$)

Specification	Coefficient (host)	SE	95% CI	p-value	log-likelihood	McFadden pseudo- R^2	Notes
Ordered logit, pooled (+ log_pop + log_gdp)	-13.73	2.03	[-17.71, -9.75]	< 0.001	-173.85	0.487	Negative = better rank
Ordered logit, pref FE + year FE (+ log_pop + log_gdp)	-49.56	172.45	[-387.56, +288.43]	0.77	-111.19	0.672	Separation-driven degeneracy
Top-1 logit, pooled (+ log_pop + log_gdp)	+9.09	2.13	[+4.92, +13.27]	< 0.001 (exact 1.9×10^{-5})	-12.75	0.707	
Top-1 logit, pref FE + year FE (+ log_pop + log_gdp)	+1,877.3	(undefined)	(undefined)	(undefined)	-0.00	1.000	Complete separation (pseudo- $R^2 = 1$ is a signature, not a fit)

Prefecture-clustered standard errors (47 clusters, Liang-Zeger sandwich) for the pooled specifications; Fisher-information for the fixed-effect specifications, where prefecture FE renders further clustering mechanically redundant and where SE degeneracy is itself a diagnostic signature of complete separation. All specifications above include log-population and log-GDP as covariates by construction; the “pooled” label denotes the absence of prefecture and year fixed effects, not the absence of covariates. The two pooled rows above are numerically identical (coefficient and clustered SE) to the M3 (+ log GDP) row of Table 4. Fixed-effect specifications are reported for transparency but not interpreted as coefficient estimates; see Methods.

Table 3. Funahashi (2016) replication (2003-2011 tennou, n = 423)

Specification	Coefficient (host)	SE (pref clusters)	p-value	R ²
Base (pref FE + year FE, no controls)	+1,575.45	57.58	< 0.001	0.94
Pooled OLS, no fixed effects	+1,733.29	138.26	< 0.001	0.33
Extended (2003-2012, +1 year)	+1,604.36	60.08	< 0.001	0.94
Funahashi (2016) reported base	+1,674.65	—	(n.r.)	0.86

47 prefecture clusters. Our replication omits Funahashi's socioeconomic controls (see Limitations); the 5.9% gap versus Funahashi's headline coefficient is within the expected range for the omitted-controls difference.

Table 4. Csurilla-style staged specification (2012-2022 tennou, n = 423)

Stage	Model	Host coefficient (top-1 / rank)	SE (top-1 / rank)	95% CI (top-1 / rank)	p (top-1 / rank)	log-likelihood (top-1 / rank)	pseudo-R ² (top-1 / rank)
M1	Host only	+5.61 / -6.11	1.09 / 1.21	[+3.47, +7.75] / [-8.48, -3.74]	< 0.001 / < 0.001	-23.50 / -308.20	0.461 / 0.091
M2	+ log population	+10.90 / -13.17	3.35 / 2.01	[+4.33, +17.47] / [-17.11, -9.23]	0.0011 / < 0.001	-14.83 / -176.49	0.660 / 0.479
M3	+ log GDP	+9.09 / -13.73	2.13 / 2.03	[+4.92, +13.27] / [-17.71, -9.75]	< 0.001 / < 0.001	-12.75 / -173.85	0.707 / 0.487
M4	+ prefecture FE	+498 / -116	NaN / NaN	(undefined) / (undefined)	(degenerate) / (degenerate)	-3.43 / -112.51	0.921 / 0.668
M5	+ year FE	+1,877 / -49.56	NaN / 172.45	(undefined) / [-387.56, +288.43]	(degenerate) / 0.77	-0.00 / -111.19	1.000 / 0.672

Standard errors are prefecture-clustered (47 clusters, Liang-Zeger sandwich) for M1-M3 and Fisher-information for M4/M5 (where prefecture FE renders clustering mechanically redundant and complete separation drives SEs to degeneracy). The M3 row is numerically identical (coefficient and clustered SE) to the corresponding pooled row of Table 2. The attenuation Csurilla & Fertő (2023) documented at the Olympics (host coefficient shrinks by ~45% from M1 to fully adjusted) is inverted at the Kokutai: M1 → M3 raises the top-1 coefficient by ~62% and the ordered-rank coefficient by ~125%. See Discussion for interpretation.

Table 4b. Csurilla-style staged specification with Tokyo excluded (Finding #12 sensitivity)

Stage	Model	Host coefficient (top-1 / rank)	SE (top-1 / rank)	p (top-1 / rank)	log-likelihood (top-1 / rank)	pseudo-R ² (top-1 / rank)
M1	Host only	+14.61 / -18.27	0.75 / 1.14	< 0.001 / < 0.001	-5.29 / -264.36	0.804 / 0.124
M2	+ log population	+17.40 / -37.19	0.78 / 1.68	< 0.001 / < 0.001	-5.29 / -162.47	0.805 / 0.461
M3	+ log GDP	+17.66 / -24.00	0.86 / 1.98	< 0.001 / < 0.001	-5.15 / -162.46	0.810 / 0.461
M4	+ prefecture FE	+122.8 / -98.0	NaN / NaN	(degenerate) / (degenerate)	-0.00 / -107.00	1.000 / 0.645
M5	+ year FE	+67.3 / -68.3	NaN / NaN	(degenerate) / (degenerate)	-0.00 / -105.81	1.000 / 0.649

$n = 414$ (= 423 total – 9 Tokyo prefecture-year cells; equivalent to 46 prefectures $\times$ 9 years, with 2013 Tokyo's own self-host year also removed so the effective host count drops from 9 to 8). Prefecture-clustered SE on 46 clusters for M1–M3; Fisher-information for M4/M5. All three non-host top-1 wins in the full-sample panel (2016, 2017, 2022) are Tokyo wins and are removed by the exclusion, producing a complete separation on top-1 (8 of 8 remaining hosts won; 0 of 406 non-hosts won). The near-doubling of the M3 host coefficient (top-1 +9.09 $\rightarrow$ +17.66; ordered rank $-13.73 \rightarrow -24.00$ vs. Table 4) is the direct quantitative confirmation of the Csurilla-reverse mechanism discussed in Results and Discussion — Tokyo's non-host championship mass was suppressing the estimated host coefficient in the full sample. This is offered as a mechanism diagnostic, not as a preferred point estimate; the separation regime drives Fisher-information SEs downward and inflates the coefficient beyond the full-sample identifying variation, and Table 4 (with Tokyo retained) remains the primary quantitative claim.

Table 5. 2024-2025 cross-section: Balmer2003 host $\times$ subjective interaction (n = 6,991; 2 treated prefectures out of 47)

Specification	Coefficient (host)	Coefficient (host $\times$ subj)	SE (int)	95% CI (int)	p (int)	Cluster SE
Baseline (score, sport FE + year FE), cluster-robust p	+16.60	+16.68	7.72	[+1.55, +31.81]	0.031	47 pref clusters
Baseline, wild-cluster bootstrap (Rademacher, B = 999, restricted null)	+16.60	+16.68	7.72	(see Baseline row; bootstrap CI not tabulated)	0.155	47 pref clusters (2 treated)
Diagnostic only — not identified: Three-way (obj/ semi/subj interactions)	+12.99	+20.31	NaN	(rank-deficient)	(rank- deficient)	—
Log(1 + score) baseline, cluster-robust p	+0.467	+0.293	0.170	[-0.040, +0.626] (crosses 0)	0.084 (marginal)	47 pref clusters

47 prefecture clusters (Saga and Shiga are the two treated clusters), robust SE throughout. Sport ($n = 40-41$) and year ($n = 2$) fixed effects included in all specifications. Because only 2 of 47 prefectures are treated in this cross-section, we complement the cluster-robust p with a wild-cluster bootstrap p (Cameron, Gelbach & Miller, 2008); the bootstrap correction moves the interaction p from 0.031 to 0.155 and dominates the cluster-robust p for inference under the few-treated-clusters regime (Cameron & Miller, 2015). The three-way spec is reported descriptively only owing to rank deficiency from sport-and-category multicollinearity.

Table 6. Descriptive host-boost by sport category (2024-2025, per-sport score)

Category	Mean non-host	Mean host	Cells (non-host)	Cells (host)	Host boost (points)
Objective ($n = 17$ sports)	26.03	43.63	2,920	65	+17.60
Semi-subjective / team ($n = 13$ sports)	20.85	47.03	2,199	48	+26.18
Subjective ($n = 11$ sports)	19.33	57.24	1,721	38	+37.91

Monotonic increase in raw host boost across the three categories matches the direction predicted by Balmer2003. The subjective category's +37.9 is the largest of the three and is roughly $2.15\times$ the objective category's +17.6, consistent with the model-implied doubling in Table 5.

Table 7. Cross-national context: host-scoring institutional arrangements

System	Host score bonus	Empirical host effect	Reference
Japan Kokutai (present study)	None (no institutional bonus in JSPO scoring)	+33 pts subj / +17 pts obj per sport	Present study
Korea national comprehensive sports	20% score bonus (10% from 2001, 20% from 2010)	Institutionally amplified	Kim (2025), Munhwa Ilbo ⁸
China National Games	(institutional bonus not documented in accessible sources)	Qualitatively described (5 characteristics: high efficacy, gradual increase, delayed effect, exclusivity, double-edgedness)	Lan & Yu (2012) ¹¹
Summer Olympics	None	Attenuates ~45% under population + GDP + country FE	Csurilla & Ferto ⁹ (2023)

The Japanese case is the only one of the four in which a large host effect emerges without any institutional score bonus and without an obvious socioeconomic driver.

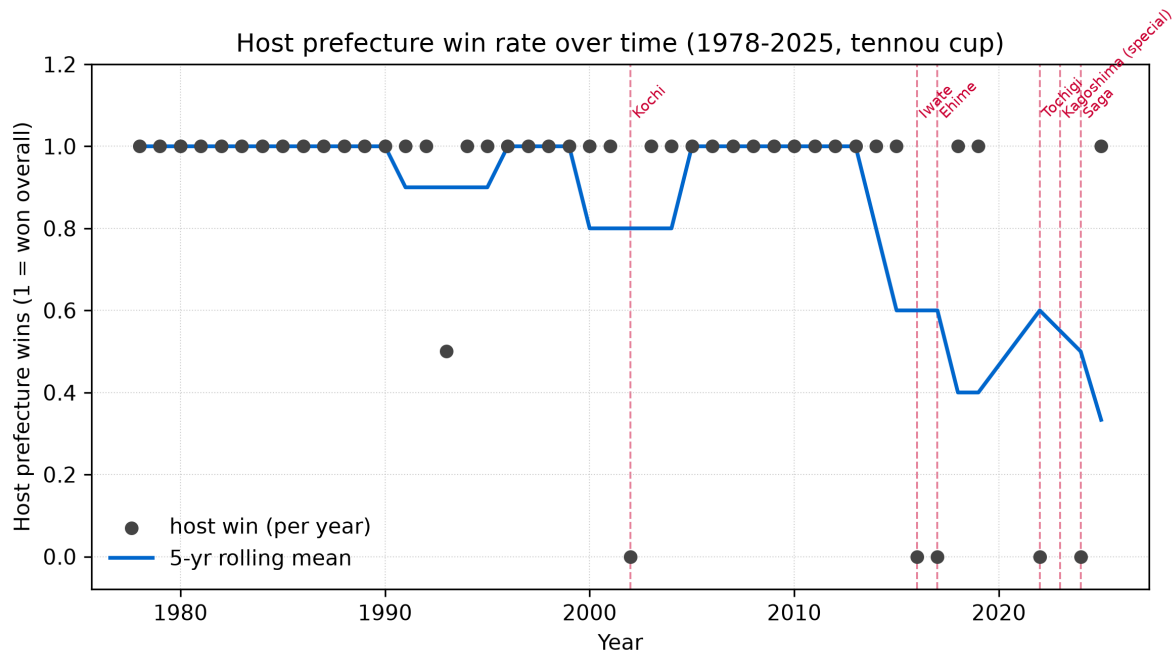


Figure 1. Host-championship rate at the Kokutai emperor's cup, 1948-2025. Each point is one edition (with the 2020/2021 cancelled and 2023 special-edition years handled as noted in the figure). Six host-loss years (2002 Kochi, 2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga) are highlighted in red. The two post-2016 host-loss clusters (2016-17 and 2022-24) are separated by host wins in 2018 Fukui and 2019 Ibaraki. *Note on the 1993 point at $y \approx 0.5$.* The 48th edition (1993) was co-hosted by Kagawa and Tokushima — the only jointly hosted edition in the 1948-2025 series. Kagawa finished 1st (host_win = 1) and Tokushima finished 2nd (host_win = 0), so the two-prefecture host mean is $(1 + 0) / 2 = 0.5$. This is a historical feature of the schedule, not a data artifact.

Host bias by sport category (2024 Saga + 2025 Shiga, tennou+kougou pooled;
error bars = 95% CI from prefecture-clustered SE, 47 clusters)

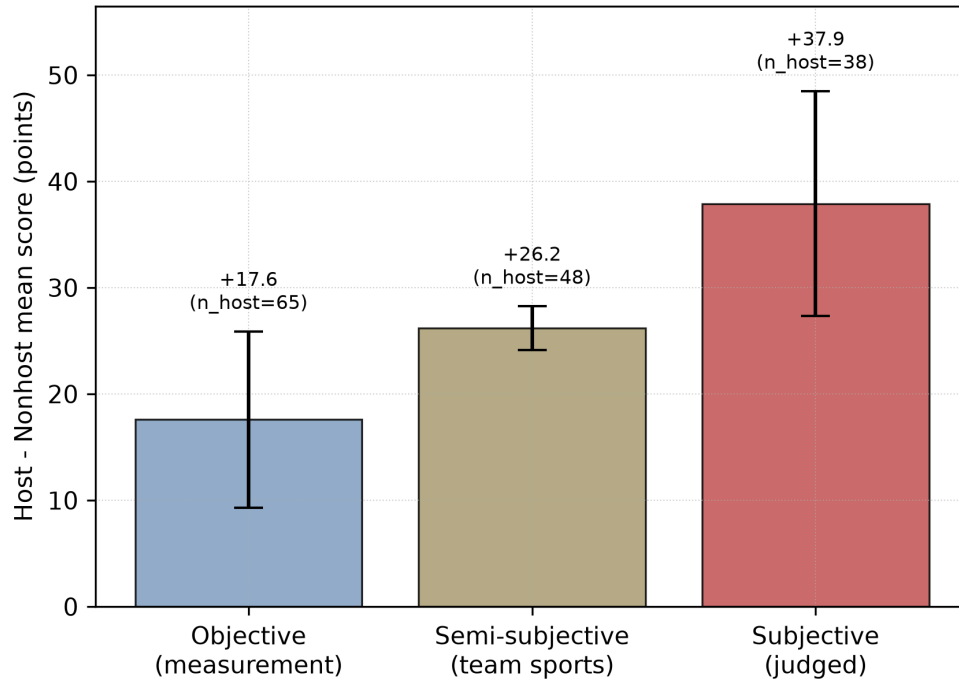


Figure 2. Host-versus-non-host per-sport score gap, by Balmer2003 sport category, in the 2024-2025 stacked cross-section ($n = 6,991$). Bars show the mean host boost in points; error bars show 95% confidence intervals ($1.96 \times$ prefecture-clustered SE, 47 clusters, per-category `'score ~ is_host'` OLS). Objective sports ($n = 17$, $n_host_cells = 65$): +17.6 points. Semi-subjective / team sports ($n = 13$, $n_host_cells = 48$): +26.2 points. Subjective sports ($n = 11$, $n_host_cells = 38$): +37.9 points. The monotonic ordering matches the Balmer2003 prediction.

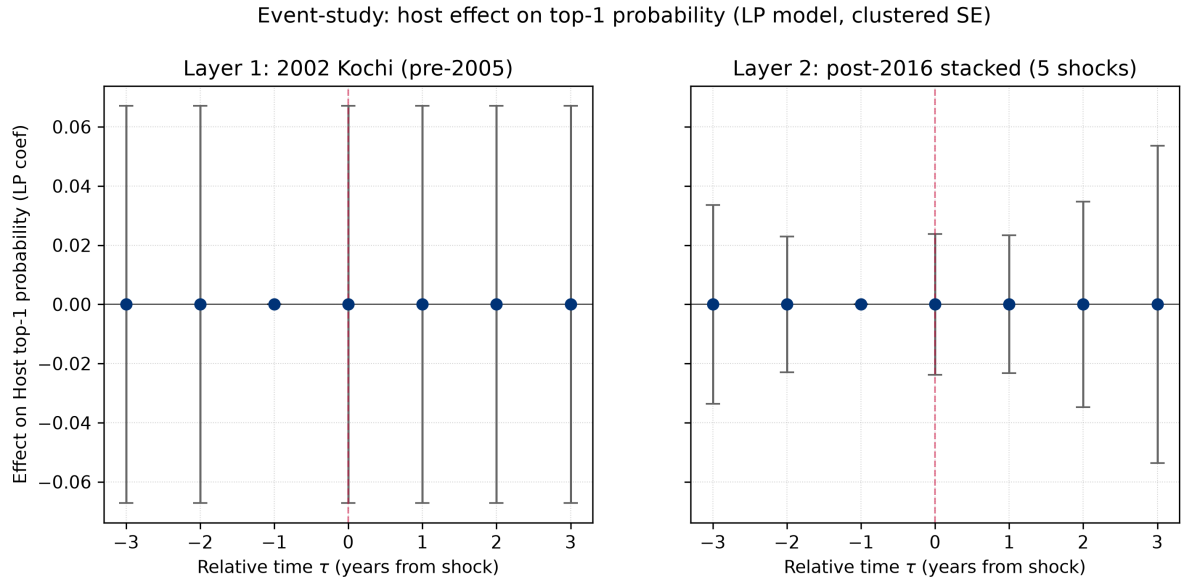


Figure 3. Two-layer event-study τ -coefficient trails on top-1 (championship) status. Layer 1: 2002 Kochi single host-loss event (pre-2005 **furusato*-athlete-rule* regime). Layer 2: post-2016 stacked five host-loss events (2016 Iwate, 2017 Ehime, 2022 Tochigi, 2023 Kagoshima special, 2024 Saga). The identifying variation is thin (see Results); the figure is included as a design check rather than as an independent identifying analysis.

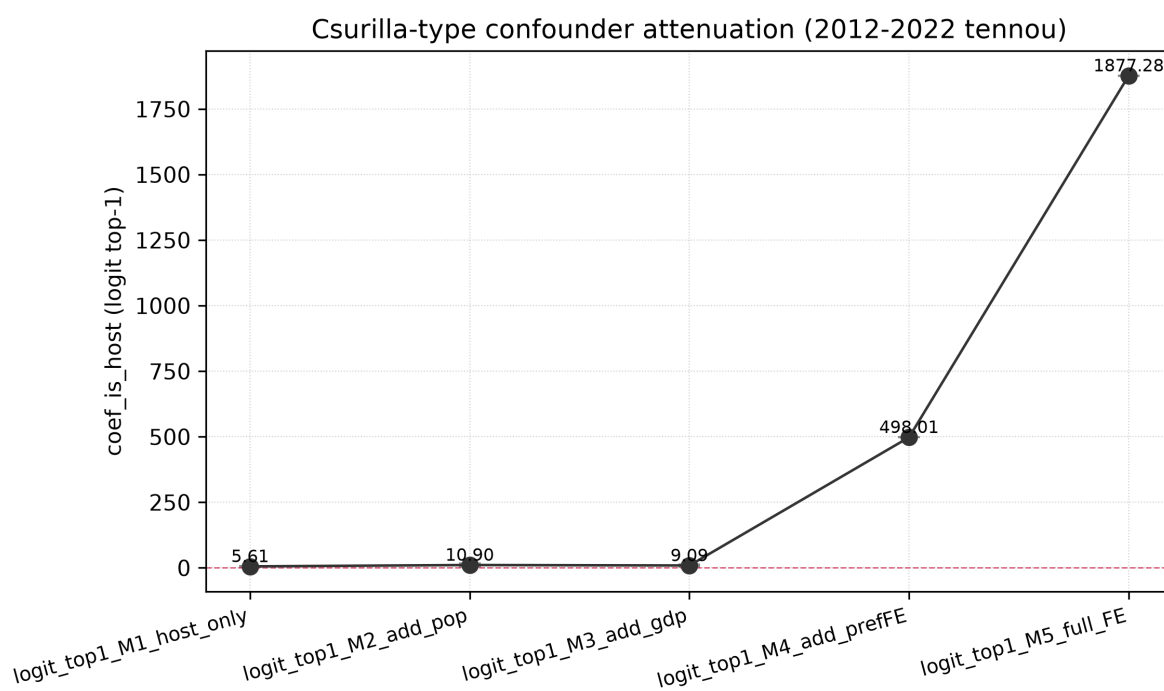


Figure 4. Csurilla-style M1→M5 staged specification, top-1 outcome. The host coefficient grows in absolute magnitude as population and GDP controls are added (M1 = +5.61, M2 = +10.90, M3 = +9.09), the opposite of the ~45% attenuation Csurilla & Fertő (2023) report for the Olympics. M4 and M5 exhibit separation-driven degeneracy and are shown truncated.

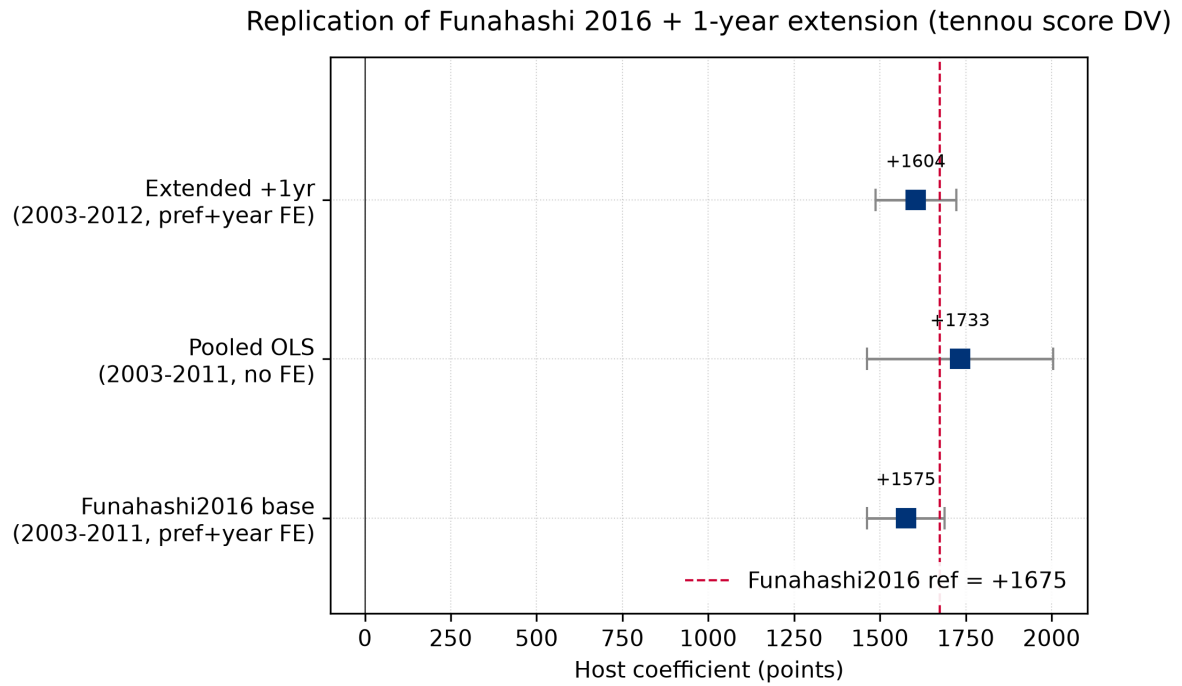


Figure 5. Funahashi (2016) replication, three specifications on the 2003-2011 emperor's-cup total-score panel. ``funahashi_base`` (prefecture FE + year FE, no controls) recovers +1,575.45; ``pooled_no_fe`` gives +1,733.29; ``extended_2003_2012`` (+1 year) gives +1,604.36. Funahashi's headline coefficient of +1,674.65 is shown as a dashed reference line; error bars are 95% confidence intervals ($1.96 \times \text{SE}$). The ``pooled_no_fe`` specification lies within one SE of the reference; ``funahashi_base`` and ``extended_2003_2012`` lie within 1.96 SE (95% CI) of the reference.